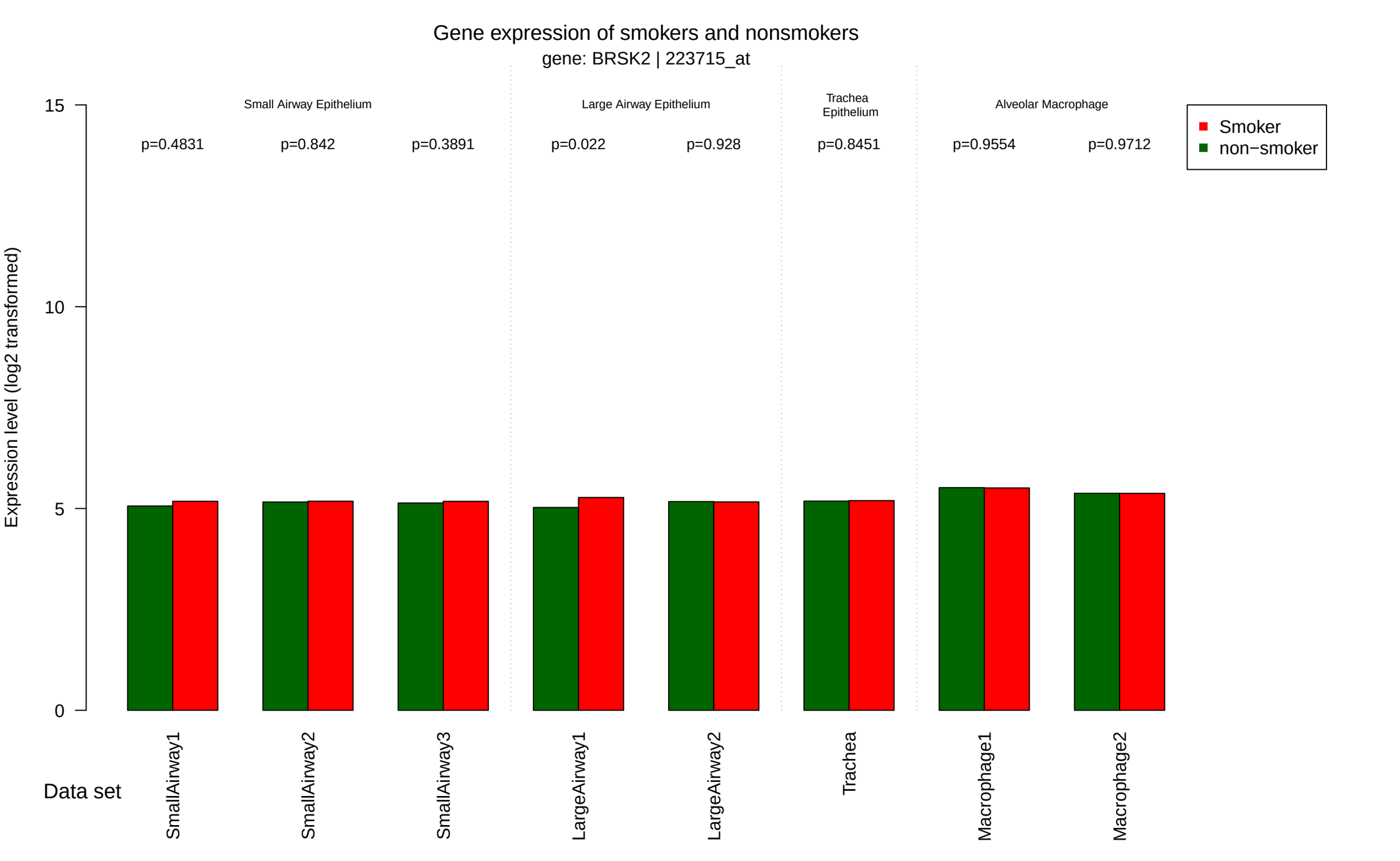
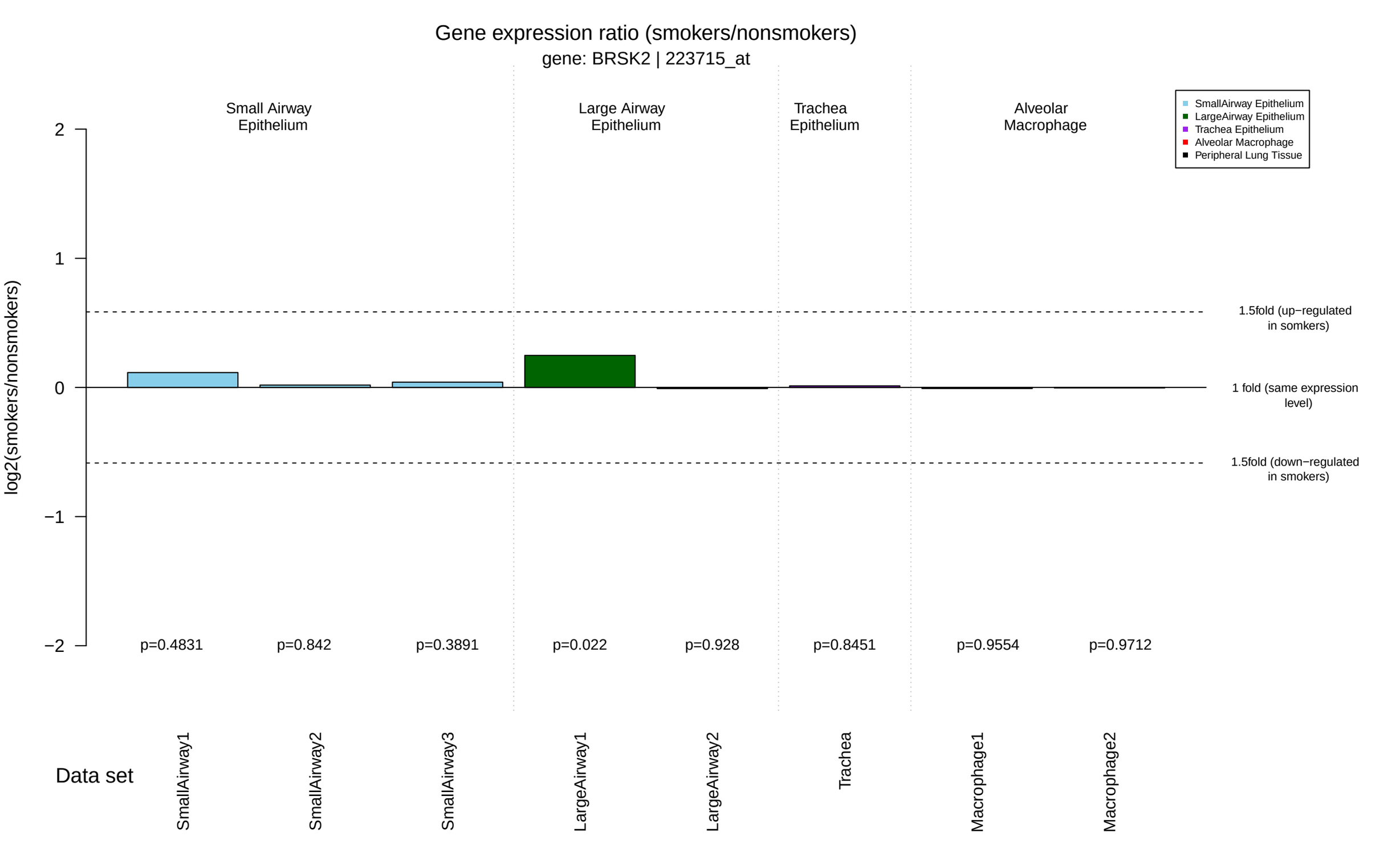
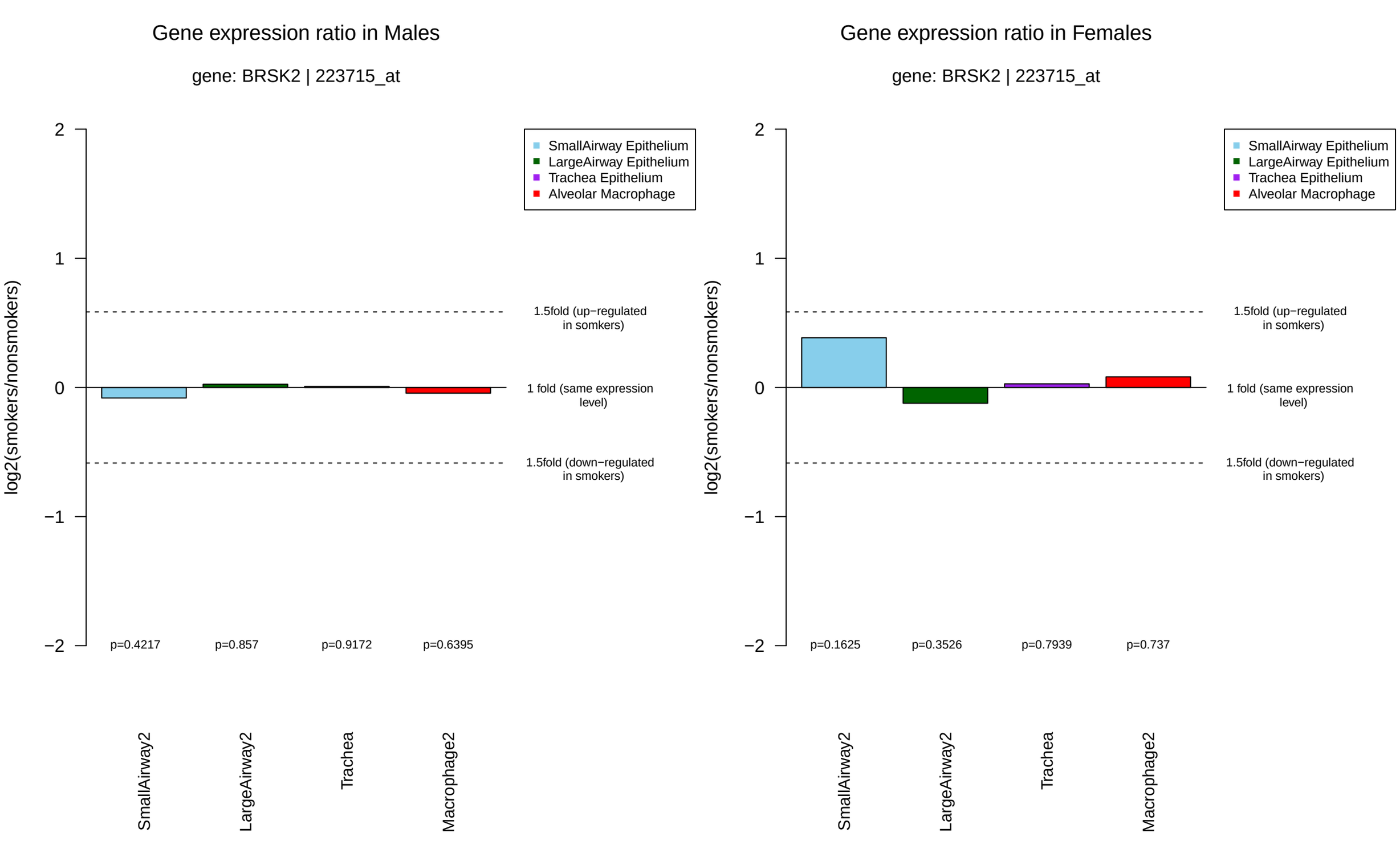
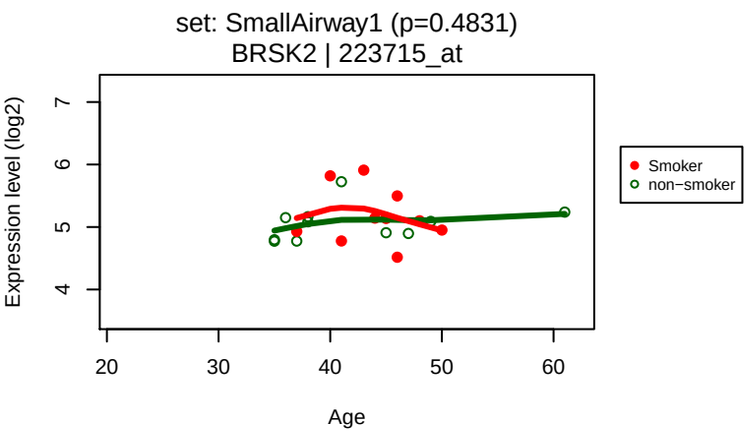
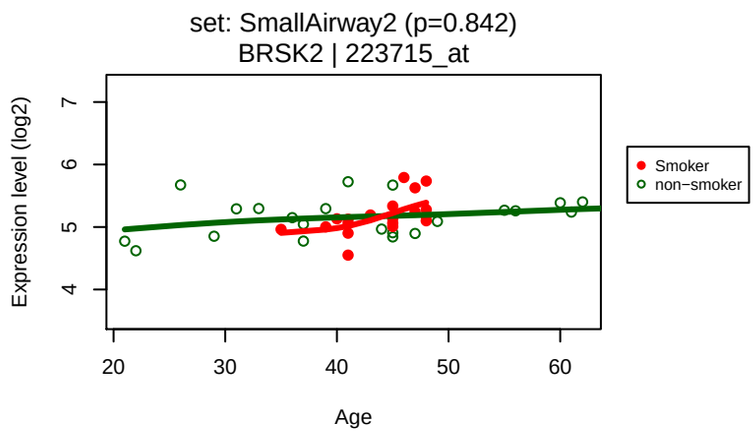
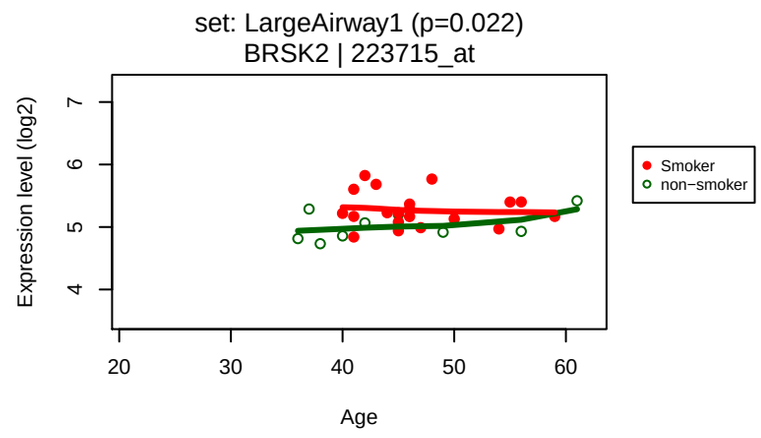

Gene expression of smokers and nonsmokers
gene: BRSK2 | 223715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4831
p=0.842
p=0.3891
p=0.022
p=0.928
p=0.8451
p=0.9554
p=0.9712
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

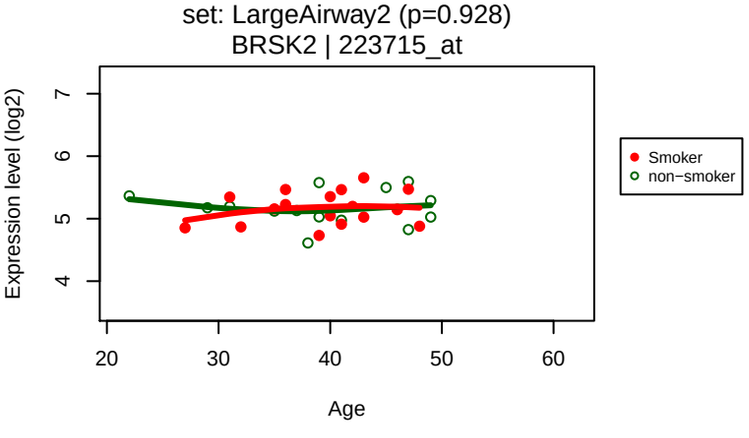

Gene expression ratio (smokers/nonsmokers)
gene: BRSK2 | 223715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4831
p=0.842
p=0.3891
p=0.022
p=0.928
p=0.8451
p=0.9554
p=0.9712
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

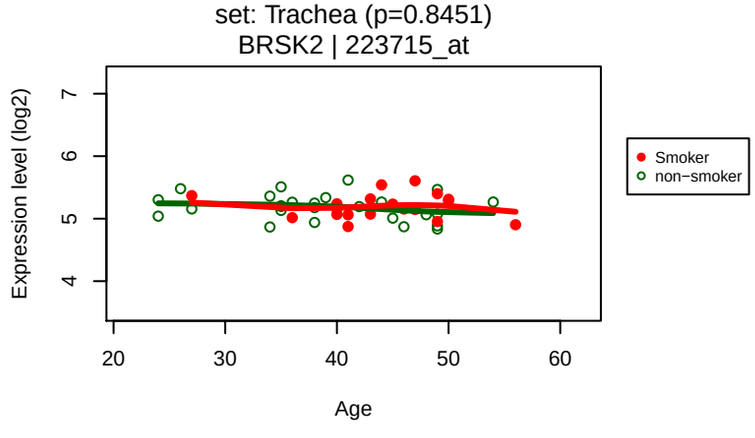

Gene expression ratio in Males
gene: BRSK2 | 223715_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4217
p=0.857
p=0.9172
p=0.6395
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRSK2 | 223715_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1625
p=0.3526
p=0.7939
p=0.737
SmallAirway2
LargeAirway2
Trachea
Macrophage2

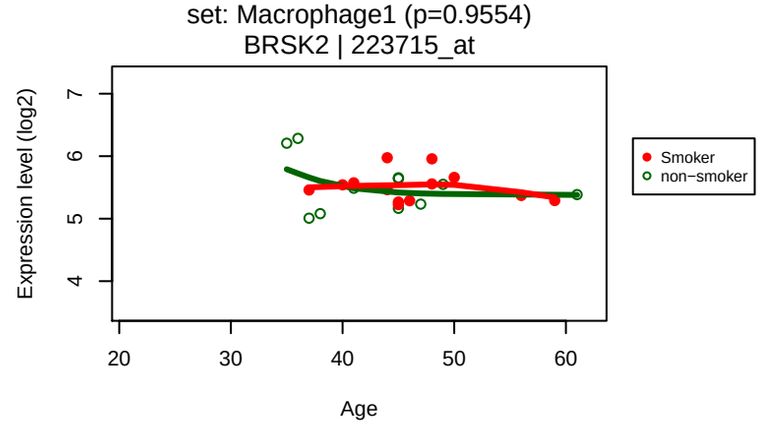

set: SmallAirway1 (p=0.4831)
BRSK2 | 223715_at
Expression level (log2)
Age
Smoker
non−smoker

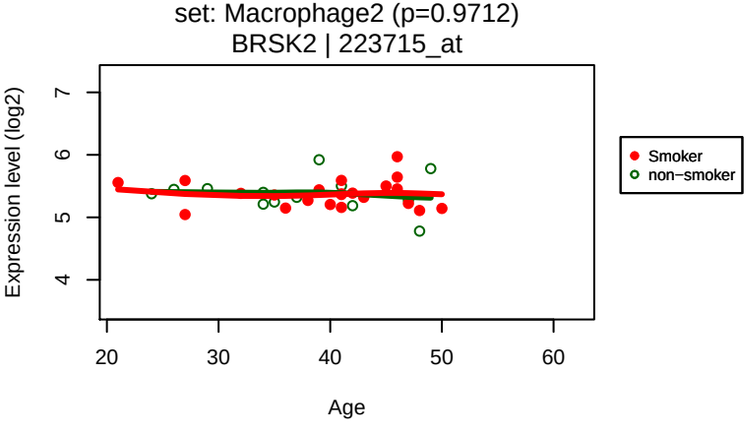

set: SmallAirway2 (p=0.842)
BRSK2 | 223715_at
Expression level (log2)
Age
Smoker
non−smoker

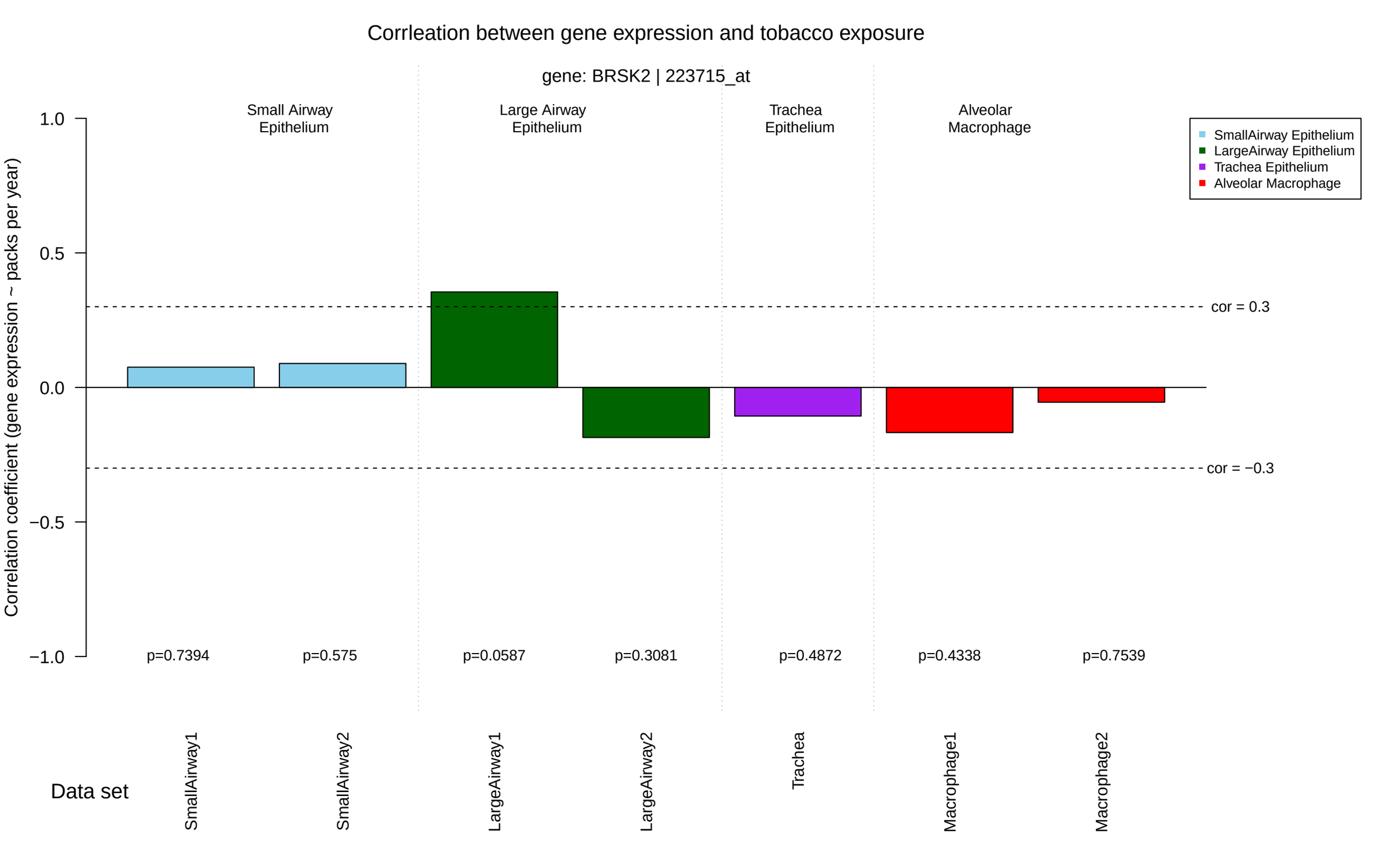

set: LargeAirway1 (p=0.022)
BRSK2 | 223715_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.928)
BRSK2 | 223715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8451)
BRSK2 | 223715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9554)
BRSK2 | 223715_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9712)
BRSK2 | 223715_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRSK2 | 223715_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7394
p=0.575
p=0.0587
p=0.3081
p=0.4872
p=0.4338
p=0.7539
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2